7 KUKA-interface standard API

7.1 Mission related APIs

7.1.1 Dispatch mission (Upstream system -> KUKA system)

API info

API Name	submitMission
API Description	Dispatch mission to KUKA.AMR Fleet.
API URL	http://[IP:Port]/interfaces/api/amr/submitMission
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Public request parameters

Name	Type	Max Length	Required	Default Value	Description
orgId	String		T		Organization ID (or factory code, supplier code).
requestId	String	32	T		Request ID. Process request idempotency.
missionCode	String		T		Mission code. Mission code should be unique.
missionType	String		T		Mission type: • Lift AMR Mission: RACK_MOVE • Conveyor AMR Mission: ROLLER_MOVE • Carton Transfer Unit Mission: PICKER_MOVE • Forklift AMR Mission: FORKLIFT_MOVE • Move Mission: MOVE • Composite AMR Mission: ROBOTICS_MOVE

Name	Type	Max Length	Required	Default Value	Description
viewBoardType	String		F		Task board displays task types, defined by user.
robotType	String		F		Robot function type: • Lift AMR: LIFT • Conveyor AMR: ROLLER • Carton Transfer Unit: PICKER • Forklift AMR: FORKLIFT • HCS Box Robot: HCSBOX • Composite Robot: COMPOSITE In principle, it should not be empty; if it is empty, it will be inferred through specific robot or container type codes.
robotModels	List<String>		F		Robot type code. Support specifying multiple robot models, one of which will be selected to execute the mission.
robotIds	List<String>		F		Robot ID. Support specifying multiple robots, one of which will be selected to execute the mission.
priority	Integer		F	1	Job Priority: 1 (highest) to 99 (lowest). Default: 1. Valid when dispatching batch tasks.
containerModelCode	String		F		Container model code.
containerCode	String		F		Container code.
templateCode	String		F		Workflow external code. If the mission is dispatched using the workflow external code, the workflow corresponding to the workflow external code will be created without the need to set missionData.

Name	Type	Max Length	Required	Default Value	Description
lockRobotAfterFinish	Boolean		F	false	Whether the robot needs to maintain the task locking status after the workflow ends. Mostly used in scenarios where a new workflow still needs to use the robot for carry after the current workflow ends.
unlockRobotId	String		F		Unlock the current robot from the previous workflow. If the previous task set lockRobotAfterFinish = true, this field is required.
unlockMissionCode	String		F		The mission code to lock the current robot. Used to verify whether the unlocking is correct.
idleNode	String		F		After the workflow is completed, the robot will go to the idleNode.
missionData	List<?>		T		The list of nodes included in the current workflow, refer to the following section for specific fields. If templateCode is set, this field can be omitted.

7.1.1.1 Lift AMR mission (missionType=RACK_MOVE)

Request parameters

Name	Type	Required	Default Value	Description
sequence	Integer	T		The default serial number is 1, which is the starting node of the starting shelf, and subsequent nodes increase in sequence.
position	String	T		Mission target node or area.
type	String	T		Mission node type: • Node: NODE_POINT • Area: NODE_AREA
actionType	String	F		Actions to be performed at nodes: • Lift up the container: UP_CONTAINER • Lower the container: DOWN_CONTAINER No need to be set if the AMR only moves and does not need to handle a container.

Name	Type	Required	Default Value	Description
putDown	Boolean	F	false	**Deprecated** Decide whether the shelf needs to be lowered at the current node.
passStrategy	String	F	AUTO	Resume strategy at current node: • Automatically resume: AUTO • Manually resume: MANUAL
waitingMillis	Integer	F	0	The waiting time before automatically triggering the next node task. Required when passStrategy=true.

Request body example

```
{
  "orgId": "UNIVERSAL",
  "requestId": "request202309250001",
  "missionCode": "mission202309250001",
  "missionType": "RACK_MOVE",
  "viewBoardType": "",
  "robotModels": [
   "KMP600I"
  ],
  "robotIds": [
    "14"
  ],
  "robotType": "LIFT",
  "priority": 1,
  "containerModelCode": "10001",
  "containerCode": "1000002",
  "templateCode": "",
  "lockRobotAfterFinish": false,
  "unlockRobotId": "",
  "unlockMissionCode": "",
```

```
  "idleNode": "A000000013",
  "missionData": [
    {
      "sequence": 1,
      "position": "M001-A001-45",
      "type": "NODE_POINT",
      "actionType": "UP_CONTAINER",
      "passStrategy": "AUTO",
      "waitingMillis": 0
    },
    {
      "sequence": 2,
      "position": "M001-A001-40",
      "type": "NODE_POINT",
      "actionType": "DOWN_CONTAINER",
      "passStrategy": "AUTO",
      "waitingMillis": 0
    }
  ]
}
```

Response body example

```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```

7.1.1.2 Conveyor AMR mission (missionType=ROLLER_MOVE)

Request parameters

Name	Type	Required	Default Value	Description
sequence	Integer	T		The default serial number is 1, which is the starting node of the starting shelf, and subsequent nodes increase in sequence.
position	String	T		Mission target node or area.
type	String	T		Mission node type: • Node: NODE_POINT • Area: NODE_AREA
binCode	String	T		Container code carried by the conveyor AMR.
rollerLevel	Integer	F	1	Level of roller AMR layers. Required for multi-layer conveyor AMR.
deviceCode	String	F		Device code. When a point corresponds to multiple devices, this value needs to be passed.
actionType	String	F		Actions to be performed at the point: • Pick up the container: ROLLER_RECEIVE • Place the container: ROLLER_SEND Optional when manually picking or placing the container.
actionConfirm	Boolean	F	false	Whether confirmation is required before picking/placing. Generally used for interaction with conveyor lines.
actionInform	Boolean	F	false	Whether notification is required after picking/placing. Generally used for interaction with conveyor lines.

Request body example

```
{
  "orgId": "UNIVERSAL",
  "requestId": "request202309250002",
  "missionCode": "mission202309250002",
  "missionType": "ROLLER_MOVE",
```

```
  "viewBoardType": "",
  "robotModels": [
     "KMP600I-1-T2"
  ],
 "robotIds": [
     "15"
  ],
  "robotType": "ROLLER",
  "priority": 1,
  "templateCode": "",
  "lockRobotAfterFinish": false,
  "unlockRobotId": "",
  "unlockMissionCode": "",
  "idleNode": "A000000013",
  "missionData": [
     {
        "sequence": 1,
        "position": "M001-A001-49",
        "type": "NODE_POINT",
        "actionType": "ROLLER_RECEIVE",
        "binCode": "1000002",
      "rollerLevel": 1,
        "deviceCode": "bea8f740-5dcc-11ee-91ba-c7b4e0c01550",
        "actionConfirm": false,
        "actionInform": false
     },
     {
        "sequence": 2,
        "position": "M001-A001-40",
        "type": "NODE_POINT",
```

```
        "actionType": "ROLLER_SEND",
        "binCode": "1000002",
        "rollerLevel": 1,
        "deviceCode": "bea8f740-5dcc-11ee-91ba-c7b4e0c01550",
        "actionConfirm": false,
            "actionInform": false
      }
   ]
}
```

Response body example

```
{
   "data": null,
   "code": "0",
   "message": null,
   "success": true
}
```

7.1.1.3 Carton transfer unit mission (missionType=PICKER_MOVE)

Request parameters

Name	Type	Required	Default Value	Description
sequence	Integer	T		The default serial number is 1, which is the starting node of the starting shelf, and subsequent nodes increase in sequence.
startPosition	String	T		Start node of the carton.
startSlotCode	String	T		Slot code at the start node.
endPosition	String	T		End node of the carton.
endSlotCode	String	T		Slot code at the end node.
binCode	String	T		Bin code.
takeActionConfirm	Boolean	F	false	Whether to confirm after picking up the container. Generally used for interaction with the conveyor line.

Name	Type	Required	Default Value	Description
takeActionInform	Boolean	F	false	Whether to inform before picking up the container. Generally used for interaction with the conveyor line.
putActionConfirm	Boolean	F	false	Whether to confirm after placing the container. Generally used for interaction with the conveyor line.
putActionInform	Boolean	F	false	Whether to inform before placing the container. Generally used for interaction with the conveyor line.

Request body example

```
{
  "orgId": "",
  "requestId": "",
  "missionCode": "",
  "missionType": "",
  "viewBoardType": "",
  "robotModels": [],
  "robotIds": [],
  "robotType": "",
  "priority": 0,
  "containerModelCode": "",
  "containerCode": "",
  "templateCode": "",
  "lockRobotAfterFinish": true,
  "unlockRobotId": "",
  "unlockMissionCode": "",
  "idleNode": "",
  "missionData": [
    {
      "sequence": 1,
```

```
        "binCode": "box-001",
        "startPosition": "",
        "startSlotCode": "",
        "takeActionConfirm": false,
        "takeActionInform": false,
        "endPosition": "",
        "endSlotCode": "",
        "putActionConfirm": false,
        "putActionInform": false
    },
    {
        "sequence": 2,
        "binCode": "box-002",
        "startPosition": "",
        "startSlotCode": "",
        "takeActionConfirm": false,
            "takeActionInform": false,
        "endPosition": "",
        "endSlotCode": "",
        "putActionConfirm": false,
            "putActionInform": false
    },
    {
        "sequence": 3,
        "binCode": "box-003",
        "startPosition": "",
        "startSlotCode": "",
        "takeActionConfirm": false,
            "takeActionInform": false,
        "endPosition": "",
```

```
        "endSlotCode": "",
        "putActionConfirm": false,
             "putActionInform": false
      }
   ]
}
```

Response body example

```
{
   "data": null,
   "code": "0",
   "message": null,
   "success": true
}
```

7.1.1.4 Forklift AMR mission (missionType=FORKLIFT_MOVE)

Request parameters

Name	Type	Required	Default Value	Description
sequence	Integer	T		The default serial number is 1, which is the starting node of the starting shelf, and subsequent nodes increase in sequence.
position	String	T		Mission target node or area.
type	String	T	NODE_POINT	Position type: • NODE_POINT • NODE_AREA • NODE_SLOT
stackNumber	Integer	F	0	Number of layers forked. Indicates the number of layers already stacked at the node.
actionConfirm	Boolean	F	false	Whether to confirm before forking or placing the container. Used in KUKA.AMR Fleet 2.15.0 and later versions.
actionInform	Boolean	F	false	Whether to inform after forking or placing the container. Used in KUKA.AMR Fleet 2.15.0 and later versions.

Request body example


```
{
  "orgId": "9001",
  "requestId": "request2023030800001",
  "missionCode": "mission2023030800001",
  "missionType": "FORKLIFT_MOVE",
  "viewBoardType": "W01",
  "robotType": "FORKLIFT",
  "robotModels": [],
  "robotIds": [],
  "priority": 1,
  "containerModelCode": "",
  "containerCode": "",
  "templateCode": "",
  "lockRobotAfterFinish": false,
  "unlockRobotId": "",
  "unlockMissionCode": "",
  "idleNode": "",
  "missionData": [
    {
      "sequence": 1,
      "position": "",
      "stackNumber": 0,
      "actionConfirm": false
      "actionInform": false
    },
    {
      "sequence": 2,
      "position": "",
      "stackNumber": 0,
```

```
        "actionConfirm": false

        "actionInform": false

      }

   ]

}
```

Response body example

```
{

   "data": null,

   "code": "0",

   "message": null,

   "success": true

}
```

7.1.1.5 Move mission (missionType=MOVE)

Request parameters

Name	Type	Required	Default Value	Description
sequence	Integer	T		The default serial number is 1, which is the starting node of the starting shelf, and subsequent nodes increase in sequence.
position	String	T		Mission target node or area.
type	String	T		Mission node type: • Node: NODE_POINT • Area: NODE_AREA

Name	Type	Required	Default Value	Description
passStrategy	String	F	AUTO	Resume strategy at current node: • Automatically resume: AUTO • Manually resume: MANUAL
waitingMillis	Integer	F	0	The waiting time before automatically triggering the next node task. Required when passStrategy=true.

Request body example


```
{
  "orgId": "UNIVERSAL",
  "requestId": "request202309250005",
  "missionCode": "mission202309250005",
  "missionType": "MOVE",
  "viewBoardType": "",
  "robotModels": [
    "KMP600I"
  ],
  "robotIds": [
    "14"
  ],
  "robotType": "LIFT",
  "priority": 1,
  "templateCode": "",
  "lockRobotAfterFinish": false,
  "unlockRobotId": "",
  "unlockMissionCode": "",
  "missionData": [
```

```
    {
      "sequence": 1,
      "position": "M001-A001-26",
      "type": "NODE_POINT",
      "passStrategy": " MANUAL",
      "waitingMillis": 0
    },
    {
      "sequence": 2,
      "position": "M001-A001-31",
      "type": "NODE_POINT",
      "passStrategy": " MANUAL",
      "waitingMillis": 0
    }
  ]
}
```

Response body example

```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```

7.1.1.6 Composite AMR mission (missionType=ROBOTICS_MOVE)

Request parameters

Name	Type	Required	Default Value	Description
sequence	Integer	T		The default serial number is 1, which is the starting node of the starting shelf, and subsequent nodes increase in sequence.
position	String	T		Mission target node or area.
type	String	T		Mission node type: • Node: NODE_POINT • Area: NODE_AREA
applicationName	String	T		Application name.
params	String	F		Robot arm parameters (JSON).
passStrategy	String	F	AUTO	Resume strategy at current node: • Automatically resume: AUTO • Manually resume: MANUAL
waitingMillis	Integer	F	0	The waiting time before automatically triggering the next node task. Required when passStrategy=true.

Response body example


```
{
  "orgId": "UNIVERSAL",
  "requestId": "request202311220001",
  "missionCode": "mission202311220001",
  "missionType": "ROBOTICS_MOVE",
  "viewBoardType": "",
  "robotModels": [
    "KMR-IISY"
  ],
  "robotIds": [
    "15"
  ],
  "robotType": "COMPOSITE",
  "priority": 1,
```

```
  "containerModelCode": "10001",

  "containerCode": "1000002",

  "templateCode": "",

  "lockRobotAfterFinish": false,

  "unlockRobotId": "",

  "unlockMissionCode": "",

  "idleNode": "A000000014",

  "missionData": [

    {

      "sequence": 1,

      "position": "M001-A001-45",

      "type": "NODE_POINT",

      "applicationName ": “app”,

          "params": “{\"key1\": \"value1\",\"key2\":
\"value2\"}”,

      "passStrategy": "AUTO",

      "waitingMillis": 0

    },

    {

      "sequence": 2,

      "position": "M001-A001-40",

      "type": "NODE_POINT",

      "applicationName ": “app”,

        "params": “{\"key1\": \"value1\",\"key2\":
\"value2\"}”,

      "passStrategy": "AUTO",

     "waitingMillis": 0

    }

  ]

}
```

Response body example

```
{

   "data": null,

   "code": "0",

   "message": null,

   "success": true

}
```

7.1.2 Cancel mission (Upstream system -> KUKA system)

API info

API Name	missionCancel
API Description	Cancel the mission.
API URL	http://[IP:Port]/interfaces/api/amr/missionCancel
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Max Length	Required	Default Value	Description	
requestId	String	32	T		Request ID, UUID32.	
missionCode	String		F		Mission code.	One of the three options must be filled in, preferably using the missionCode. The node is used in certain specific scenarios.
containerCode	String		F		Container code.	
position	String		F		Node UUID of the mission.	

Name	Type	Max Length	Re-quired	Default Value	Description
cancelMode	String		T	FORCE	Cancel Mode: • FORCE: Force cancel Immediately terminate the current mission. • NORMAL: Normal cancel Does not cancel the task that the robot is executing, and waits for the robot to complete the current task before canceling the workflow. • REDIRECT_END: Redirect to the end node of the workflow. Does not cancel the task that the robot is executing, and waits for the robot to complete the current task. If the target node does not require lowering the rack, the robot carries the rack to the end node; otherwise, the empty robot moves to the end node. • REDIRECT_START: Redirect to the start node of the workflow. Does not cancel the task that the robot is executing, and waits for the robot to complete the current task. If the target node does not require lowering the rack, the robot carries the rack back to the start node; otherwise, the empty robot moves back to the start node. Currently, only tasks for lift robots support all strategies, while other types of robots support FORCE only. However, it still depends on whether the robot supports cancellation.
reason	String		F		Reason to cancel.

Request body example

```
{

   "requestId": "request202309250006",
```

```
  "missionCode": "mission202309250004",
  "containerCode": "",
  "position": "",
  "cancelMode": "FORCE",
  "reason": ""
}
```

Response body example

```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```

7.1.3 Resume workflow (Upstream system -> KUKA system)

API info

API Name	operationFeedback
API Description	When the passStrategy parameter of a step is set to MANUAL, resume the job by calling this API.
API URL	http://[IP:Port]/interfaces/api/amr/operationFeedback
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Max Length	Required	Description	
requestId	String	32	T	Request ID, UUID32.	
missionCode	String		F	Mission code of the running workflow.	One of the three options must be filled in.
containerCode	String		F	Container code of the running workflow.	
position	String		F	The current node of the workflow.	

Request body example

```
{
  "requestId": "request202309250007",
  "containerCode": "",
  "missionCode": "mission202309250005",
  "position": ""
}
```

Response body example

```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```

7.1.4 Retry workflow (Upstream system -> KUKA system)

API info

API Name	retryMission
API Description	Retry the executing robot task or device task of the job.
API URL	http://[IP:Port]/interfaces/api/amr/retryMission
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Max Length	Required	Description	
taskCode	String		F	Job code of the running workflow.	One of the three options must be filled in.
containerCode	String		F	Container code of the running workflow.	
robotId	String		F	Robot ID of the running workflow.	

Request body example

```
{

  "taskCode": "task202309250007"

}
```

Response body example

```
{

"data": null,

"code": "0",

"message":"Success",

"success": true

}
```

7.1.5 Query jobs (Upstream system -> KUKA system)

API info

API Name	jobQuery
API Description	Query jobs in the job board.
API URL	http://[IP:Port]/interfaces/api/amr/jobQuery
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Default Value	Description
workflowId	Long	F		Workflow ID.
containerCode	String	F		Container code.
jobCode	String	F		Job code. Same with the mission code.
status	String	F		Job status: • Created: 10 • Executing: 20 • Waiting: 25 • Cancelling: 28 • Complete: 30 • Cancelled: 31 • Manual complete: 35 • Warning: 50 • Startup error: 60
robotId	String	F		Robot ID
targetCellCode	String	F		Target node code of the running task in the workflow.
workflowName	String	F		Name of the workflow configuration.
workflowCode	String	F		Workflow code of the workflow configuration.
maps	List<String>	F		Map code where the workflow is running.
createUsername	String	F		Operator.

Name	Type	Required	Default Value	Description
sourceValue	Integer	F		Source: • Interface: 2 • PDA: 3 • Device: 4 • MLS: 5 • Fleet application: 6 • Workflow event: 7
limit	Integer	F	10	Number of jobs in the response, default 10.

Response parameters

Name	Type	Description
jobCode	String	Job code. Same with the mission code.
workflowId	Long	Workflow ID.
containerCode	String	Container code.
robotId	String	Robot ID.
status	Integer	Job status: • Created: 10 • Executing: 20 • Waiting: 25 • Cancelling: 28 • Complete: 30 • Cancelled: 31 • Manual complete: 35 • Warning: 50 • Startup error: 60
workflowName	String	Name of the workflow configuration. Name could be empty when the workflow is generated by upstream system.
workflowCode	String	Workflow code of the workflow configuration. Workflow code could be empty when the workflow is generated by upstream system.
workflowPriority	Integer	Workflow priority.
mapCode	String	Map code.
targetCellCode	String	Target node code of the running task in the workflow.
beginCellCode	String	Start node code of the running task in the workflow.
targetCellCodeForeign	String	Foreign code of target node code of the running task in the workflow.
beginCellCodeForeign	String	Foreign code of start node code of the running task in the workflow.

Name	Type	Description
finalNodeCode	String	End node of the workflow. Area code for the area node and node code for the map node.
warnFlag	Integer	Warn flag: • Normal: 0 • Warning: 1
warnCode	String	Warn code.
completeTime	String	Workflow completion time (yyyy-MM-dd HH:mm:ss).
spendTime	Integer	Workflow time spent (second).
createUsername	String	Operator.
createTime	String	Workflow creation time (yyyy-MM-dd HH:mm:ss).
source	String	Source: • INTERFACE • PDA • DEVICE • MLS • SELF • EVENT
materialsInfo	String	Material information. Required when users need material management of containers.

Request body example

For reference only. Fill in the parameters as needed, and all fields are optional.

```
{

        "containerCode": "C001",

        "createUsername": "admin",

        "jobCode": "T000096284",

        "limit": 10,

        "maps": ["TEST"],

        "robotId": "1",

        "sourceValue": 6,

        "status": 20,

        "targetCellCode": "TEST-1-90",

        "workflowCode": " W000000587",

        "workflowId": 100218,
```

```
        "workflowName": "Carry01"

}
```

Response body example

```
{

"data": [

     {

        "jobCode": "T000096284",

        "workflowId": 100218,

        "containerCode": "C001",

        "robotId":"1",

        "status": 20,

        "workflowName": "Carry01",

        "workflowCode": " W000000587",

        "workflowPriority": 1,

        "mapCode": "TEST",

        "targetCellCode": "TEST-1-90",

        "beginCellCode": "TEST-1-80",

        "targetCellCodeForeign": "DROPPOINT",

        "beginCellCodeForeign": "PICKPOINT",

        "finalNodeCode": "TEST-1-90",

        "warnFlag": 0,

        "warnCode": null

        "completeTime": null,

        "spendTime": null,

        "createUsername": "admin",

        "createTime": "2025-01-10 16:01:42",

        "source": "SELF",

        "materialsInfo": "-"

     }  ],
```

```
  "code": "0",

  "message": null,

  "success": true

}
```

7.1.6 Pause mission (Upstream system -> KUKA system)

API info

API Name	pauseMission
API Description	Pause the mission by the job code or the robot ID.
API URL	http://[IP:Port]/interfaces/api/amr/pauseMission
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description	
robotId	String	F	The robot to pause.	One of the two options must be filled in.
missionCode	String	F	Mission code of the running workflow to pause.	

Request body example

```
Pause by robot:

{

  "robotId": "2"

}

Pause by mission:

{

  "missionCode": "mission202309250005"
```

```
}
```

Response body example

```
{
  "data": "",
  "code": "0",
  "message": "lang.wcs.common.ok",
  "success": true
}
```

7.1.7 Recover Mission (Upstream System -> KUKA System)

API info

API Name	recoverMission
API Description	Recover the mission by the job code or the robot ID..
API URL	http://[IP:Port]/interfaces/api/amr/recoverMission
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description	
robotId	String	F	The robot to recover.	One of the two options must be filled in.
missionCode	String	F	Mission code of the running workflow to recover.	

Request body example

```
Recover by robot:
```

```
{
   "robotId": "2"
}
Recover by mission:
{
   "missionCode": "mission202309250005"
}
```

Response body example

```
{
   "data": "",
   "code": "0",
   "message": "lang.wcs.common.ok",
   "success": true
}
```

7.1.8 Mission status callback (KUKA system -> Upstream system)

API info

API Name	missionStateCallback
API Description	Mission status callback.
API URL	http://[IP:Port]/interfaces/api/amr/missionStateCallback
HTTP Method	POST

Request parameters

Name	Type	Max Length	Required	Description
missionCode	String	32	T	Mission code. Same with the job code of the workflow.
viewBoardType	String		F	View board type.
containerCode	String		F	Container code.
currentPosition	String		F	Current node code of the mission.
slotCode	String		F	Current slot code.
robotId	String		F	Robot that is executing the mission.

Name	Type	Max Length	Required	Description
missionStatus	String		T	Mission status: • MOVE_BEGIN • ARRIVED • UP_CONTAINER • DOWN_CONTAINER • ROLLER_RECEIVE • ROLLER_SEND • PICKER_RECEIVE • PICKER_SEND • FORK_UP • FORK_DOWN • WAITFEEDBACK • COMPLETED • CANCELED
message	String		F	Description message.
missionData			F	Custom information object that needs to be reported.

Request body example

```
{

  "missionCode": "mission202309250005",

  "viewBoardType": "",

  "slotCode": "",

  "robotId": "14",

  "containerCode": "1000002",

  "currentPosition": "M001-A001-31",

  "missionStatus": "MOVE_BEGIN",

  "message": "",

  "missionData": {}

}
```

7.2 Container related APIs

7.2.1 Insert container (Upstream system -> KUKA system)

API info

API Name	containerIn
API Description	Insert the container.
API URL	http://[IP:Port]/interfaces/api/amr/containerIn
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Max Length	Required	Default Value	Description
requestId	String	32	T		Request ID, UUID32.
containerType	String		F		Container type: • RACK • BIN
containerModelCode	String		F		Container model code. Required when isNew = true.
containerCode	String		T		Container code.
enterOrientation	String	32	F		Container orientation after inserting. Special requirements for the orientation when inserting the container.
position	String		T		Node code where the container is inserted.
isNew	Boolean		F	false	Automatically create the container before inserting.
containerValidationCode	String		F		Appointed validation code when creating the container. When isNew=true, appointed validation code.

Name	Type	Max Length	Re-quired	Default Value	Description
withDefaultValidationCode	Boo-lean		F		Using default validation code when creating the container(the same with containerCode). When isNew=true, set the container validation code same with containerCode.

Request body example (Inert)

```
{
  "requestId": "request202309250008",
  "containerCode": "10",
  "position": "M001-A001-31",
  "containerModelCode": "",
  "enterOrientation": "",
  "isNew": false
}
```

Request body example (Add and insert with appointed validation code)

```
{
  "requestId": "request202309250008",
  "containerCode": "10",
  "position": "M001-A001-31",
  "containerModelCode": "1",
  "enterOrientation": "0",
  "isNew": true,
  "containerValidationCode": "ID:10",
  "withDefaultValidationCode": false
}
```

Request body example (Add and insert with default validation code)

```
{
  "requestId": "request202309250008",
```

```
  "containerCode": "10",

  "position": "M001-A001-31",

  "containerModelCode": "1",

  "enterOrientation": "0",

  "isNew": true,

  "containerValidationCode": "",

  "withDefaultValidationCode": true

}
```

Response body example

```
{

  "data": null,

  "code": "0",

  "message": null,

  "success": true

}
```

7.2.2 Remove container (Upstream system -> KUKA system)

API info

API Name	containerOut
API Description	Remove the container.
API URL	http://[IP:Port]/interfaces/api/amr/containerOut
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Max Length	Re-quired	Default Value	Description
requestId	String	32	T		Request ID, UUID32.
containerType	String		F	RACK	Container type: RACK.
containerCode	String		T		Container code.
position	String		F		Node code of the container.
isDelete	Boolean		F	false	Delete container after removing the container.

Request body example

```
{
  "requestId": "request202309250009",
  "containerType": "",
  "containerCode": "10",
  "position": "M001-A001-31",
  "isDelete": false
}
```

Response body example

```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```

7.2.3 Update container (Supported: update position and empty full status) (Upstream system -> KUKA system)

API info

API Name	updateContainer
API Description	Update the container.
API URL	http://[IP:Port]/interfaces/api/amr/updateContainer
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Max Length	Required	Description	
requestId	String	32	T	Request ID, UUID32.	
containerType	String		F	Container type: RACK.	
containerCode	String		T	Container code.	
originPosition	String		F	Origin node code of the container.	
targetPosition	String		F	Target node code of the container.	At least one of the fields must be filled in.
emptyStatus	String		F	Container empty/full status: • EMPTY • FULL	
reason	String		F	Reason to update.	

Request body example

```
{

  "requestId": "request202309250010",

  "containerType": "BUCKET",

  "containerCode": "10",

  "originPosition": "M001-A001-31",

  "targetPosition": "M001-A001-30",

  "emptyStatus": "EMPTY",

  "reason": ""

}
```

Response body example

```
{

  "data": null,
```

```
  "code": "0",

  "message": null,

  "success": true

}
```

7.2.4 Query container model codes (Upstream system -> KUKA system)

API info

API Name	queryAllContainerModelCode
API Description	Query all container models.
API URL	http://[IP:Port]/interfaces/api/amr/queryAllContainerModelCode
HTTP Method	GET

Headers

Param Name	Param Value	Required	Example	Notes
Authorization	${accessToken}	T		Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Response body example

```
{

  "code": "0",

  "message": null,

  "success": true,

  "data": [

    "10001"

  ]

}
```

7.2.5 Query area codes by container model (Upstream system -> KUKA system)

API info

API Name	queryAreaCodeForContainerModel
API Description	Query areas that support the current container model.
API URL	http://[IP:Port]/interfaces/api/amr/queryAreaCodeForContainerModel
HTTP Method	GET

Headers

Param Name	Param Value	Required	Notes
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
containerModelCode	String	T	Container model code.
noContainerFirst	Boolean	F	Select container operation: • true (Move the container into the idle node of an area) • false (Move the container out of the area)

Request URL example

```
http://[IP:Port]/interfaces/api/amr/queryAreaCodeForContainerModel?containerModelCode=10001& noContainerFirst=false
```

Response body example

```
{
  "code": "0",
  "message": "",
  "success": true,
  "data": [
    "areaCode1",
    "areaCode2",
    "areaCode3"
```

```
    ]
}
```

7.2.6 Query containers (Only including containers with IN/INSERTED status) (Upstream system -> KUKA system)

API info

API Name	containerQuery
API Description	Retrieve only inserted containers by criteria.
API URL	http://[IP:Port]/interfaces/api/amr/containerQuery
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Default Value	Description
containerCode	String	F		Container code.
nodeCode	String	F		Node code.
areaCode	String	F		Area code.
containerModelCode	String	F		Container model code.
emptyFullStatus	Integer	F	2	Container empty/full status: • EMPTY: 0 • FULL: 1 • ALL: 2 Empty/full status is valid only if the area code exists.

Response parameters

Name	Type	Description
containerCode	String	Container code.
nodeCode	String	Node code.

Name	Type	Description
inMapStatus	Integer	In map status: • In map: 1 • Off map: 0
containerModelCode	String	Container model code.
emptyFullStatus	String	Container empty/full status: • EMPTY: 0 • FULL: 1
isCarry	Integer	Lift status: • Drop: 0 • Lift: 1
orientation	String	Container orientation.
containerCheckCode	String	Container check code.
mapCode	String	Map code.
districtCode	String	Map district code.

Request body example

```
{
  "nodeCode": "M001-A001-30",
  "containerModelCode": "10001",
  "containerCode": "10",
  "areaCode": "A000000014",
  "emptyFullStatus": 2
}
```

Response body example

```
{
  "data": [
    {
      "containerCode": "10",
      "nodeCode": "",
      "orientation": "0.0",
      "containerModelCode": "10001",
      "emptyFullStatus": 0,
      "inMapStatus": 1,
```

```
        "isCarry": 0,
        "containerCheckCode": "10",
        "mapCode": "ABC",
        "districtCode": "1"
      }
  ],
  "code": "0",
  "message": null,
  "success": true
}
```


7.2.7 Query containers (Both IN/INSERTED and OUT/REMOVED status) (Upstream system -> KUKA system)

API info

API Name	containerQueryAll
API Description	Retrieve containers by criteria, including both inserted and removed, matching the UI representation.
API URL	http://[IP:Port]/interfaces/api/amr/containerQueryAll
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
containerCode	String	F	Container code.
nodeCode	String	F	Node code.
areaCode	String	F	Area code.
mapCode	String	F	Map code.
districtCode	String	F	District code.
containerModelCode	String	F	Container model code.

Name	Type	Required	Description
emptyFullStatus	Integer	F	Container empty/full status: • EMPTY: 0 • FULL: 1
inMapStatus	Integer	F	IN/OUT map status: • IN: 1 • OUT: 0

Response parameters

Name	Type	Description
containerCode	String	Container code.
nodeCode	String	Node code.
inMapStatus	Integer	In map status: • In map: 1 • Off map: 0
containerModelCode	String	Container model code.
emptyFullStatus	String	Container empty/full status: • EMPTY: 0 • FULL: 1
isCarry	Integer	Lift status: • Drop: 0 • Lift: 1
orientation	String	Container orientation.
containerCheckCode	String	Container check code.
mapCode	String	Map code.
districtCode	String	Map district code.

Request body example (Query all container in map ABC)

```
{

  "mapCode": "ABC"

}
```

Response body example

```
{

  "data": [

    {

      "containerCode": "10",

      "nodeCode": "1-1",

      "orientation": "0.0",
```

```
        "containerModelCode": "10001",
        "emptyFullStatus": 0,
        "inMapStatus": 1,
        "isCarry": 0,
        "containerCheckCode": "10",
        "mapCode": "ABC",
        "districtCode": "1"
},
{
        "containerCode": "20",
        "nodeCode": "1-1",
        "orientation": "0.0",
        "containerModelCode": "10001",
        "emptyFullStatus": 0,
        "inMapStatus": 0,
        "isCarry": 0,
        "containerCheckCode": "",
        "mapCode": "ABC",
        "districtCode": "1"
    }
  ],
 "code": "0",
  "message": null,
  "success": true
}
```

7.3 Robot module

7.3.1 Query robot by node UUID or node code (Upstream system -> KUKA system)

API info

API Name	queryRobByNodeUuidOrForeignCode
API Description	Query a robot by node UUID or node foreign code.
API URL	http://[IP:Port]/interfaces/api/amr/queryRobByNodeUuidOrForeignCode
HTTP Method	GET

Headers

Param Name	Param Value	Required	Notes
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
nodeCode	String	T	Node UUID or node foreign code. If querying multiple robots, the user can pass multiple node parameters separated by commas.

Response parameters

Name	Type	Description
robotId	String	Robot ID.
robotType	String	Robot type code.
mapCode	String	Map code.
floorNumber	String	Floor number.
buildingCode	String	Building code.
containerCode	String	Container code that is carried by the robot.
status	Integer	Status: • Departure: 1 • Offline: 2 • Idle: 3 • Executing: 4 • Charging: 5 • Updating: 6 • Abnormal: 7

Name	Type	Description
occupyStatus	Integer	Occupy status: • Occupied: 1 • Idle: 0
batteryLevel	Double	Battery level (%).
nodeCode	String	Node code.
X	String	Robot current location Y (millimeters).
Y	String	Robot current location X (millimeters).
robotOrientation	String	Robot orientation.
missionCode	String	Mission code. Mission code dispatched by the upstream system
liftStatus	Integer	Robot lift status: • Lift: 1
reliability	Integer	Location reliability: • Unreliable: 0 • Reliable: 1
runTime	String	Run time.
karOsVersion	String	KUKA.AMR OS version.
mileage	String	Running mileage.
leftMotorTemperature	String	Left motor temperature.
rightMotorTemperature	String	Right motor temperature.
liftMotorTemperature	String	Lift motor temperature.
rotateMotorTemperature	String	Rotate motor temperature.
rotateTimes	Integer	Rotate times.
liftTimes	Integer	Lift times.
nodeForeignCode	String	Foreign code of the node where the robot locates. This field will be populated when querying robots by node foreign code.

Request example

- Example 1:

```
http://[IP:Port]/interfaces/api/amr/
queryRobByNodeUuidOrForeignCode?nodeCode= startNode1,
startNode2
```

- Example 2:

```
http://[IP:Port]/interfaces/api/amr/
queryRobByNodeUuidOrForeignCode?nodeCode= test01-5-91
```

Response body example

- Example 1:

```
{
```

```
"data": [
  {
    "robotId": "925",
    "robotType": "KMP600I",
    "mapCode": "test01",
    "floorNumber": "5",
    "buildingCode": "test",
    "containerCode": "",
    "status": 3,
    "occupyStatus": 0,
    "batteryLevel": 1,
    "nodeCode": "test01-5-64",
    "x": "3000.0",
    "y": "15000.0",
    "robotOrientation": "90.0",
    "missionCode": null,
    "liftStatus": 0,
    "reliability ": 1,
    "runTime ": "6745",
    "karOsVersion ": "",
    "mileage ": "0.0",
    "leftMotorTemperature ": "0.0",
    "rightMotorTemperature ": "0.0",
    "liftMotorTemperature ": "0.0",
    "rotateMotorTemperature ": "0.0",
    "rotateTimes ": 0,
    "liftTimes ": 0,
    "nodeForeignCode": " startNode1"
  },
  {
```

```
        "robotId": "80211",
        "robotType": "KMP600I",
        "mapCode": "test01",
        "floorNumber": "5",
        "buildingCode": "test",
        "containerCode": "",
        "status": 3,
        "occupyStatus": 0,
        "batteryLevel": 1,
        "nodeCode": "test01-5-74",
        "x": "1200.0",
        "y": "2400.0",
        "robotOrientation": "0.0",
        "missionCode": null,
        "liftStatus": 0,
        "reliability ": 1,
        "runTime ": "null",
        "karOsVersion ": "",
        "mileage ": "0.0",
        "leftMotorTemperature ": "0.0",
        "rightMotorTemperature ": "0.0",
        "liftMotorTemperature ": "0.0",
        "rotateMotorTemperature ": "0.0",
        "rotateTimes ": 0,
        "liftTimes ": 0,
        "nodeForeignCode": " startNode2"
    }
],
"code": "0",
"message": null,
```

```
  "success": true

}
```

- Example 2:

```
{

  "data": [

    {

      "robotId": "2041",

      "robotType": "KMP600I",

      "mapCode": "test01",

      "floorNumber": "5",

      "buildingCode": "test",

      "containerCode": "",

      "status": 3,

      "occupyStatus": 0,

      "batteryLevel": 1,

      "nodeCode": "test01-5-91",

      "x": "13000.0",

      "y": "30000.0",

      "robotOrientation": "-90.0",

      "missionCode": null,

      "liftStatus": 0,

      "reliability ": 1,

      "runTime ": "6745",

      "karOsVersion ": "",

      "mileage ": "0.0",

      "leftMotorTemperature ": "0.0",

      "rightMotorTemperature ": "0.0",

      "liftMotorTemperature ": "0.0",

      "rotateMotorTemperature ": "0.0",

      "rotateTimes ": 0,
```

```
        "liftTimes ": 0,
        nodeForeignCode": ""
      },
    ],
    "code": "0",
    "message": null,
    "success": true
}
```


7.3.2 Query robots (Upstream system -> KUKA system)

API info

API Name	robotQuery
API Description	Query a robot (default to search all).
API URL	http://[IP:Port]/interfaces/api/amr/robotQuery
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description	
robotId	String	F	Robot ID.	If all parameters are empty, then query all by default. The request parameters mapCode and floorNumber must be passed together.
robotType	String	F	Robot type code.	
mapCode	String	F	Map code (If map is disabled, result is empty).	
floorNumber	String	F	Floor number (If map is disabled, result is empty).	

Response parameters

Name	Type	Description
robotId	String	Robot ID.
robotType	String	Robot type code.
mapCode	String	Map code.
floorNumber	String	Floor number.
buildingCode	String	Building code.
containerCode	String	Container code that is carried by the robot.
status	Integer	• Departure: 1 • Offline: 2 • Idle: 3 • Executing: 4 • Charging: 5 • Updating: 6 • Abnormal: 7
occupyStatus	Integer	Occupy status: • Occupied: 1 • Idle: 0
batteryLevel	Double	Battery level (%).
nodeCode	String	Node code.
X	String	Robot current location Y (millimeters).
Y	String	Robot current location X (millimeters).
robotOrientation	String	Robot orientation.
missionCode	String	Mission code. Mission code dispatched by the upstream system
liftStatus	Integer	Robot lift status: • Lift: 1
reliability	Integer	Location reliability: • Unreliable: 0 • Reliable: 1
runTime	String	Run time.
karOsVersion	String	KUKA.AMR OS version.
mileage	String	Running mileage.
leftMotorTemperature	String	Left motor temperature.
rightMotorTemperature	String	Right motor temperature.
liftMotorTemperature	String	Lift motor temperature.
rotateMotorTemperature	String	Rotate motor temperature.
rotateTimes	Integer	Rotate times.
liftTimes	Integer	Lift times.

Request body example


```
{
        "floorNumber": "A001",
        "mapCode": "M001",
        "robotId": "13",
        "robotType": "KMP600I"
}
```


Response body example


```
{
  "data": [
    {
      "robotId": "13",
      "robotType": "KMP600I",
      "mapCode": "M001",
      "floorNumber": "A001",
      "buildingCode": "W001",
      "containerCode": "",
      "status": 3,
      "occupyStatus": 0,
      "batteryLevel": 1,
      "nodeCode": "46",
      "x": "3000.0",
      "y": "15000.0",
      "robotOrientation": "90.0",
      "missionCode": null,
      "liftStatus": 0,
      "reliability": 1,
      "runTime": "6745",
      "karOsVersion: "",
      "mileage": "0.0",
```

```
        "leftMotorTemperature": "0.0",
        "rightMotorTemperature": "0.0",
        "liftMotorTemperature": "0.0",
        "rotateMotorTemperature": "0.0",
        "rotateTimes": 0,
        "liftTimes": 0,
      },
    ],
    "code": "0",
    "message": null,
    "success": true
}
```


7.3.3 Dispatch move carry task (Upstream system -> KUKA system)

API info

API Name	robotMoveCarry
API Description	Dispatch a move carry task to the robot. The function is the same as that in the Map Monitor, is directly dispatched to the Mission Manager, is only displayed in the Mission List, and is not visible on the Job Board. The robot won't be locked after the task completes.
API URL	http://[IP:Port]/interfaces/api/amr/robotMoveCarry
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
robotId	String	T	Robot ID.
containerCode	String	T	Container code.
targetNodeCode	String	T	Target node code.
missionCode	String	F	Mission code that is dispatched to the Mission Manager. Automatically generating the mission code, if the parameter is not passed.

Request body example

```
{

        "robotId": "10001",

        "containerCode": "C0002",

        "targetNodeCode": "TEST-1-45"

}
```

Response body example

```
{

   "data": "I-CARRY-1744609148201",

   "code": "0",

   "message": "OK",

   "success": true

}
```

7.3.4 Dispatch lift up container task (Upstream system -> KUKA system)

API info

API Name	robotLift
API Description	Dispatch a lift container task to the robot. The function is the same as that in the Map Monitor, is directly dispatched to the Mission Manager, is only displayed in the Mission List, and is not visible on the Job Board. The robot won't be locked after the task completes.
API URL	http://[IP:Port]/interfaces/api/amr/robotLift
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Default Value	Description
robotId	String	T		Robot ID.
containerCode	String	F		Container code.
moveLift	Integer	F	1	0: Lift up 1: Move lift
missionCode	String	F		Mission code that is dispatched to the Mission Manager. Automatically generate the mission code, if the parameter is not passed.

Request body example

```
{
  "robotId": "10001",
  "containerCode": "C0002",
  "moveLift": 0
}
```

Response body example


```
{
  "data": "I-LIFT-1744609148201",
  "code": "0",
  "message": "OK",
  "success": true
}
```


7.3.5 Dispatch drop down container task (Upstream system -> KUKA system)

API info

API Name	robotDrop
API Description	Dispatch a drop down container task to the robot. The function is the same as that in the Map Monitor, is directly dispatched to the Mission Manager, is only displayed in the Mission List, and is not visible on the Job Board. The robot won't be locked after the task completes.
API URL	http://[IP:Port]/interfaces/api/amr/robotDrop
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
robotId	String	T	Robot ID.
nodeCode	String	F	Node UUID or Node foreign code. If the value is given, the robot will execute the move-lift task. Otherwise, the robot will execute the lift-up task.
missionCode	String	F	Mission code that is dispatched to the Mission Manager. Automatically generate the mission code, if the parameter is not passed.

Request body example

```
{

  "robotId": "10001",

  "nodeCode": "TEST-1-1001"

}
```

Response body example

```
{

  "data": "I-DROP-1744609148201",

  "code": "0",

  "message": "OK",

  "success": true

}
```

7.3.6 Dispatch charge robot task

API info

API Name	chargeRobot
API Description	Dispatch a charge task to the robot. The function is the same as which in the Map Monitor, directly dispatched to the Mission Manager, only displayed in the Mission List, and not visible on the Job Board.
API URL	http://[IP:Port]/interfaces/api/amr/chargeRobot
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
robotId	String	T	Robot ID.
necessary	Integer	T	Whether to force generating the charging task. • 1: Always generate a task and queue it. • 0: Charge when below lowestLevel. Task can be directly completed if no resources are available.
targetNodeCode	String	T	Target node code.
targetLevel	Integer	T	Target battery level.
lowestLevel	Integer	F	Lowest battery level to trigger charge. Required when necessary = 0.
missionCode	String	F	Mission code that is dispatched to the Mission Manager. Automatically generate the mission code, if the parameter is not passed.

Request body example

```
{

  "lowestLevel": 5,

  "necessary": 0,
```

```
  "robotId": "2",
  "targetLevel": 90
}
```

Response body example

```
{
  "data": null,
  "code": "0",
  "message": "OK",
  "success": true
}
```

7.3.7 Dispatch move task (Upstream system -> KUKA system)

API info

API Name	robotMove
API Description	Dispatch a move task to the robot. The function is the same as that in the Map Monitor, is directly dispatched to the Mission Manager, is only displayed in the Mission List, and is not visible on the Job Board. The robot won't be locked after the task completes.
API URL	http://[IP:Port]/interfaces/api/amr/robotMove
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
robotId	String	T	Robot ID.
nodeCode	String	F	Node UUID or Node foreign code.

Name	Type	Required	Description
missionCode	String	F	Mission code that is dispatched to the Mission Manager. Automatically generate the mission code, if the parameter is not passed.

Request body example

```
{

  "robotId": "10001",

  "nodeCode": "TEST-1-1001"

}
```

Response body example

```
{

  "data": "I-MOVE-1744609148201",

  "code": "0",

  "message": "OK",

  "success": true

}
```

7.3.8 Unlock robot

API info

API Name	unlockRobot
API Description	Unlock the robot. Only for abnormal recovery scenarios. Before request, ensure that there is no corresponding executing jobs for the robot on the Job Board. Otherwise, it may cause a workflow execution error.
API URL	http://[IP:Port]/interfaces/api/amr/unlockRobot
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
robotId	String	T	Robot ID.

Request body example

```
{

   "robotId": "10001"

}
```

Response body example

```
{

   "code": "0",

   "message": "Success",

   "success": true

}
```

7.3.9 Insert the robot into the map

API info

API Name	insertRobot
API Description	Insert the robot into the map. Only support real AMR currently (distinguished from simulated AMR).
API URL	http://[IP:Port]/interfaces/api/amr/insertRobot
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
robotId	String	T	Robot ID.
cellCode	String	T	Node code to insert the robot.
synchroContainer	Integer	T	Whether to insert with the container: • 0: No • 1: Yes

Request body example

```
{

  "cellCode": "TEST-1-69",

  "robotId": "2",

  "synchroContainer": 1

}
```

Response body example

```
{

  "data": null,

  "code": "0",

  "message": "OK",

  "success": true

}
```

7.3.10 Remove the robot from the map (Upstream system -> KUKA system)

API info

API Name	removeRobot
API Description	Remove the robot from the map.
API URL	http://[IP:Port]/interfaces/api/amr/removeRobot
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Default Value	Description
robotId	String	T		Robot ID.
withContainer	Integer	F	1	Whether to remove with the container: • 0: No • 1: Yes

Request body example

```
{
        "robotId": "2",
        "withContainer": 1
}
```

Response body example

```
{
  "data": null,
  "code": "0",
  "message": "OK",
  "success": true
```

```
}
```

7.4 AMR map node and area related module

7.4.1 Query all WCS area information (Upstream system -> KUKA system)

API info

API Name	areaQuery
API Description	Query all WCS area information.
API URL	http://[IP:Port]/interfaces/api/amr/areaQuery
HTTP Method	GET

Headers

Param Name	Param Value	Required	Notes
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Response parameters

Name	Type	Description
areaName	String	Area name.
areaCode	String	Area code.
areaType	Integer	Area type: • 1: Container storage area • 3: Temporary storage area • 4: Buffer area • 5: Dense storage area • 6: Exclusive area • 7: Fork area

Response body example

```
{
  "data": [
    {
      "areaName": "container storage area",
      "areaType": 1,
       "areaCode": "1-song-1732701220548"
```

```
    },
    {
      "areaName": "temporary storage area",
      "areaType": 3,
      "areaCode": "1-song-1732701269477"
    },
    {
      "areaName": "buffer area",
      "areaType": 4,
      "areaCode": "1-song-1732701316971"
    },
    {
      "areaName": "dense storage area",
      "areaType": 5,
      "areaCode": "1-song-1732701347592"
    },
    {
      "areaName": "exclusive area",
      "areaType": 6,
      "areaCode": "1-song-1732701422092"
    },
    {
      "areaName": "fork area",
      "areaType": 7,
      "areaCode": "1-song-1732701459113"
    },
  ],
  "code": "0",
  "message": null,
  "success": true
```

```
}
```

7.4.2 Query nodes of the area (Upstream system -> KUKA system)

API info

API Name	areaNodesQuery
API Description	Query nodes of the area.
API URL	http://[IP:Port]/interfaces/api/amr/areaNodesQuery
HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
areaCodes	List<String>	T	Area codes.

Response parameters

Name	Type	Description
areaCode	String	Area code.
nodeList	List<String>	The set of external codes for the nodes. If all points in the area have no external codes, this field may be empty.

Request URL example

- Example 1

```
{
        "areaCodes": ["A000000013","A000000014"]
}
```

- Example 2

```
{
        "areaCodes": ["1-song-1732612877805"]
}
```

Response body example

- Example 1


```
{
  "data": [
    {
      "areaCode": "A000000014",
      "nodeList": [
        "27"
      ]
    } ,
    {
      "areaCode": "A000000013",
      "nodeList": [
        "46",
        "50"
      ]
    }
  ],
  "code": "0",
  "message": null,
  "success": true
}
```


- Example 2, all nodes within the area have no external codes.


```
{
  "data": [
    {
      "areaCode": "1-song-1732612877805",
      "nodeList": [
        ""
```

```
        ]
      }
  ],
  "code": "0",
  "message": null,
  "success": true
}
```


7.4.3 Query area by map node (Upstream system -> KUKA system)

API info

API Name	queryWCSAreaByMapNode
API Description	Query area by map node.
API URL	http://[IP:Port]/interfaces/api/amr/queryWCSAreaBy-MapNode
HTTP Method	GET

Headers

Param Name	Param Value	Required	Notes
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
nodeUuid	String	T	Node UUID.

Response parameters

Name	Type	Description
areaCode	String	Area code.
containerModelCode	String	Model code of containers that can be placed in the area. Not required.

Request URL example

```
http://[IP:Port]/interfaces/api/amr/queryWCSAreaByMap-
Node?nodeUuid= 1-song-44
```

Response body example

- Query the area to which the node belongs, and the area is not configured with a container model code that can be placed.


```
{
   "data": {
      "areaCode": "1-song-1732612831825",
      "containerModelCode": ""
   },
   "code": "0",
   "message": null,
   "success": true
}
```


- Query the area to which the node belongs, and the area is not configured with a container model code that can be placed.


```
{
   "data": {
      "areaCode": "1-song-1732612877805",
      "containerModelCode": "600i"
   },
   "code": "0",
   "message": null,
   "success": true
}
```


- Can not find the area to which the point belongs.


```
{
   "data": null,
   "code": "100001",
   "message": "No such node in the graph.[7788]",
   "success": false
}
```

7.4.4 Query all forbidden areas (Upstream system -> KUKA system)

API info

API Name	queryAllForbiddenAreas
API Description	Query all forbidden areas.
API URL	http://[IP:Port]/interfaces/api/amr/queryAllForbiddenAreas
HTTP Method	GET

Headers

Param Name	Param Value	Required	Notes
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Response parameters

Name	Type	Description
forbiddenAreaId	Integer	Forbidden area ID.
mapCode	String	Map code where the area locates.
floorNumber	String	District code where the area locates.
description	String	Area description.
forbiddenAreaStrategy	Integer	Area strategy: • 0-default • 1-allow AMR parking
status	Integer	Status: • 0-disabled • 1-enabled • 2-effective • 3-ineffective
enableType	Integer	Effective strategy: • 0-immediately • 1-effective according to the given time period

Response body example

```
{
  "data": [
    {
      "forbiddenAreaId": 3,
```

```
        "mapCode": "1",
        "floorNumber": "song",
        "description": "",
        "forbiddenAreaStrategy": 0,
        "status": 1,
        "enableType": 0
      },
      {
        "forbiddenAreaId": 4,
        "mapCode": "1",
        "floorNumber": "1",
        "description": "",
        "forbiddenAreaStrategy": 0,
        "status": 1,
        "enableType": 0
      },
      {
        "forbiddenAreaId": 2,
        "mapCode": "1",
        "floorNumber": "song",
        "description": "forbidden area1 of map song",
        "forbiddenAreaStrategy": 0,
        "status": 1,
        "enableType": 0
      },
    ],
    "code": "0",
    "message": null,
    "success": true
}   ],
```

```
  "code": "0",
  "message": null,
  "success": true
}
```

7.4.5 Query one forbidden area (Upstream system -> KUKA system)

API info

API Name	queryOneForbiddenArea
API Description	Query one forbidden area.
API URL	http://[IP:Port]/interfaces/api/amr/queryOneForbidde-nArea
HTTP Method	GET

Headers

Param Name	Param Value	Required	Notes
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
forbiddenAreaId	Integer	T	Forbidden area ID.

Response parameters

Name	Type	Description
forbiddenAreaId	Integer	Forbidden area ID.
mapCode	String	Map code where the area locates.
floorNumber	String	District code where the area locates.
description	String	Area description. Not required.
forbiddenAreaStrategy	Integer	Area strategy: • 0-default • 1-allow AMR parking

Name	Type	Description
status	Integer	Status: • 0-disabled • 1-enabled • 2-effective • 3-ineffective
enableType	Integer	Effective strategy: • 0-immediately • 1-effective according to the given time period

Request URL example

```
http://[IP:Port]/interfaces/api/amr/queryOneForbiddenArea?forbiddenAreaId=2
```

Response body example


```
{
  "data": {
       "forbiddenAreaId": 2,
       "mapCode": "1",
       "floorNumber": "song",
       "description": "forbidden area1 of map song",
       "forbiddenAreaStrategy": 0,
       "status": 1,
       "enableType": 0
  },
  "code": "0",
  "message": null,
  "success": true
}
```


7.4.6 Update forbidden area status (Upstream system -> KUKA system)

API info

API Name	updateForbiddenAreaStatus
API Description	Update the forbidden area status.
API URL	http://[IP:Port]/interfaces/api/amr/updateForbiddenAreaStatus

HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Default Value	Description	
forbiddenAreaId	Integer	T	0	Forbidden area ID.	Only one of forbiddenAreaId or forbiddenAreaCode can be selected as a parameter in the request body at a time.
forbiddenAreaCode	String	T		Forbidden area code.	
status	String	T		Forbidden area status: • 0-disabled • 1-enabled	

Request body example

- Example 1

```
{

    "forbiddenAreaId": 2,

    "status": "0"

}
```

- Example 2

```
{

    "forbiddenAreaCode": "map1-01-1739427001143",

    "status": "0"

}
```

Response body example

- Modify status successfully.

```
{

     "data": null,

     "code": "0",

     "message": null,

     "success": true

}
```


- Fail to modify status.

  Error: The forbidden area is already disabled (status=0). Attempting to disable it again will trigger this error. Similarly, if the area is already enabled (status=1) and you attempt to enable it again, the same error will occur.


```
{

     "data": null,

     "code": "100001",

     "message": "RCS returns
error[[D]com.kuka.rcs.bd.common.grpc.util.CommonGrpcErrorC
ode.16[can not update [ MAP_ZONE_STATUS_DISABLE ]]]",

     "success": false

}
```


- An error in passing the parameters.

  Error: Only one of forbiddenAreaId or forbiddenAreaCode can be selected as a parameter in the request body at a time.


```
{

     "data": null,

     "code": "100001",

     "message": "only one of area ID or area code can
be filled in",

     "success": false

}
```


7.4.7 Query function node (Upstream system -> KUKA system)

API info

API Name	queryFunctionNode
API Description	Query the function node.
API URL	http://[IP:Port]/interfaces/api/amr/queryFunctionNode

HTTP Method	POST

Headers

Param Name	Param Value	Required	Notes
Content-Type	application/json	T	
Authorization	${accessToken}	T	Refer to (>>> *4.7 "Interface platform" Page 131*) > **Authentication** for details on the method to obtain the accessToken.

Request parameters

Name	Type	Required	Description
functionType	Integer	T	Node function type: • 1: Workstation • 2: Shelf point • 3: Reset point • 4: Charging entry point • 5: Interaction point • 6: Avoidance point • 7: Rotation point • 8: Queuing point • 9: Check point • 10: Charging point • 11: Elevator point • 12: Box shelf point • 13: Pipeline point • 15: Voice broadcast point • 16: Mobile box shelf point • 17: Pipeline relative point • 18: Wireless charging point • 19: Connect point • 20: CTU loader node point
mapCode	String	T	Map code.
floorNumber	String	T	Floor number.
robotTypeClass	String	F	Robot type class.
containerModel	String	F	Container model code.

Response parameters

Name	Type	Description
nodeCode	String	Node code.
externalCode	String	Node external code.
functionType	Integer	Node function type.
containerModelCode	List<String>	Node supported container model code.

Name	Type	Description
robotTypeCode	List<String>	Node supported robot type code.

Request body example

```
{
  "floorNumber": "1",
  "functionType": 1,
  "mapCode": "TEST"
}
```

Response body example

```
{
  "data": [
    {
      "nodeCode": "TEST-1-53",
      "externalCode": "",
      "functionType": 1,
      "containerModelCode": [
        "model1"
      ],
      "robotTypeCode": [
        "KMP600I"
      ]
    },
    {
      "nodeCode": "TEST-1-1",
      "externalCode": "",
      "functionType": 1,
      "containerModelCode": [],
      "robotTypeCode": []
    }
  ],
```

```
  "code": "0",
  "message": "lang.wcs.common.ok",
  "success": true
}
```

7.5 Interface response description

Success response

```
{
  "code": "0",
  "message": "",
  "success": true,
  "data": ""
}
```

Failure response

```
{
  "code": "", //Error code
  "message": "", //Error description
  "success": false,
  "data": null
}
```

Common error description

Error Description	Error Code	Message
Unexpected error	100001	Unknown error.
Invalid parameter	100002	Invalid parameter.
Required param missing	100404	Miss param.
Enum value not exists	100013	Invalid enum value.
No such node code	100204	No such node.
Robot or robot type not exists	100301	Robot not exists.
Request ID is duplicate	100408	API request ID is duplicate.
Mission code is duplicate	100409	API mission code is duplicate.